HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Analyzed and troubleshot **hardware and software issues** on semiconductor manufacturing equipment in a **cleanroom environment**, resolving an average of **4 critical system faults per week**.
- Executed **factory-issued hardware and software upgrades** on complex CMP capital equipment, ensuring seamless integration and adherence to technical specifications.
- Interpreted **technical drawings, electrical schematics, and P&ID; diagrams** to diagnose and resolve intricate system performance problems, reducing troubleshooting time by **18%**.
- Managed and controlled the inventory of **spare parts** supplied to customer sites, implementing a tracking system that improved part availability by **22%**.
- Reported system performance and ongoing activities to onsite customers and local management on a **weekly basis**, maintaining clear and concise communication.
- Performed **preventive maintenance** on assigned installation equipment base, extending the operational lifespan of critical components and minimizing unscheduled downtime.
- Containerized diagnostic tools and deployment scripts using **Docker** and managed their lifecycle on **AWS EC2** instances for consistent execution across customer sites.

SKILLS

Semiconductor Equipment – CMP Systems, Cleanroom Environment, Hardware, Software, PLC

Technical Documentation – Technical Drawings, Electrical Schematics, P&ID; Drawings

Operations & Maintenance – Installation, Upgrades, Repairs, Preventive Maintenance, Troubleshooting, Root Cause Analysis

Customer & Management Reporting – System Performance Reporting, Stakeholder Communication, Onsite Support

Inventory Management – Spare Parts Management, Tracking Systems

Cloud & Containerization – AWS EC2, Docker

PROJECTS

CMP Equipment Diagnostic Tooling (HCL Technologies)

- Developed a suite of Python scripts to automate the diagnosis of common hardware and software faults on semiconductor manufacturing equipment, reducing manual diagnostic efforts by **40%**.
- Integrated diagnostic scripts with **AWS Lambda** for remote execution and log aggregation via **CloudWatch**, enabling faster response times for field service engineers.
- Created comprehensive documentation and usage guides for the diagnostic tooling, facilitating adoption by the on-site support team.

Automated System Upgrade Deployment (HCL Technologies)

- Engineered a Python-based automation framework to streamline the deployment of software upgrades for enterprise systems, cutting deployment time from 2 days to **6 hours**.
- Implemented version control and rollback capabilities using **Git**, ensuring safe and repeatable upgrade processes across multiple environments.
- Collaborated with operations teams to define upgrade procedures and perform UAT, ensuring system stability post-deployment.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)